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RESEARCH

Assessment and prediction of Water Quality Index (WQI) by seasonal key water parameters in a coastal city: application of machine learning models

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Abstract The Water Quality Index (WQI) provides comprehensive assessments in river systems; however, its calculation involves numerous water quality parameters, costly in sample collection and laboratory analysis. The study aimed to determine key water parameters and the most reliable models, considering seasonal variations in the water environment, to maximize the precision of WQI prediction by a minimal set of water parameters. Ten statistical or machine learning models were developed to predict the WQI over four seasons using water quality dataset collected in a coastal city adjacent to the Yellow Sea in China, based on which the key water parameters were identified and the variations were assessed by the Seasonal-Trend decomposition procedure based on Loess (STL). Results indicated that model performance generally improved with adding more input variables except Self-Organizing Map (SOM). Tree-based ensemble methods like Extreme Gradient Boosting (XGB) and Random Forest (RF) demonstrated the highest accuracy, particularly in winter. Nutrients (Ammonia Nitrogen (AN) and Total Phosphorus (TP)), Dissolved Oxygen (DO), and turbidity were determined as key water parameters, based on which, the prediction accuracy for Medium and Low grades was perfect while it was over 80% for the Good grade in spring and winter and dropped to around 70% in summer and autumn. Nutrient concentrations were higher at inland stations; however, it worsened at coastal stations, especially in summer. The study underscores the importance of reliable

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WQI prediction models in water quality assessment, especially when data is limited, which are crucial for managing water resources effectively.

Keywords Water Quality Index · Key water parameters · Machine learning models · Spatiotemporal variations · Seasonal-Trend decomposition procedure based on Loess

Introduction

With the rapid development of urbanization and expansion of industrial activities, massive amounts of industrial waste and urban effluents have been discharged into surface waters, leading to widespread deterioration in water quality (Zhang et al., 2024a). Additionally, polluted runoff from agricultural lands, resulting from the excessive use of pesticides and fertilizers, has exacerbated eutrophication and harmful algal blooms, posing severe threats to aquatic ecosystems and the usability of water resources (Saeed et al., 2024). Faced with these challenges, water quality evaluation is an essential step in developing effective water resource management strategies and successful implementation of pollution control measures (Wieczorek et al., 2024).

The challenges of managing water quality and mitigating pollution in the coastal cities are compounded by their unique geographical positions and concentrated economic activities. These areas typically host vital port facilities that support extensive commercial, industrial, and fishing operations (Zhang & Zhang, 2023). As a result, the water bodies in these regions are exposed to a multitude of pollution sources, including land-based sewage discharges, industrial effluents, and agricultural runoff (Xiong et al., 2023). Additionally, the impacts of extreme weather events, such as the increased influx of pollutants into rivers during heavy rainfall and storms, have exacerbated the water pollution issues (Dunn et al., 2022). Several studies have highlighted the water pollution issues in coastal cities worldwide. For instance, the Rushikulya estuary on the east coast of Odisha, India, is in a moderate to high eutrophic state due to excessive phosphorus (Gogoi et al., 2024); anthropogenic nutrient inputs are contributing to phytoplankton assemblages in the northern Gulf of Mexico (Bargu et al., 2019); and nutrient limitations are affecting phytoplankton in the Chesapeake Bay, the largest estuary in the USA (Zhang et al., 2022). In China, significant attention has been drawn to the main influence of riverine inputs on heavy metal pollution in Zhanjiang Bay in the South China Sea (Peng et al., 2024), episodic hypoxia events caused by interactions between large river discharges and ambient water in the Changjiang River Estuary in the East China Sea (Zhang et al., 2023) and acidification and eutrophication issues in Laizhou Bay, Bohai Sea (Wei et al., 2022). Similarly, previous research has highlighted the trophic status problems caused by intensive nutrient application and human activities in the Yellow Sea, which have severely deteriorated the water environment (Sun et al., 2022). However, these studies have primarily focused on Shandong Province and limited evaluation of water quality was reported in the coastal cities of Jiangsu Province, which play crucial roles in the regional economy and ecological protection of the Yellow Sea, particularly for Lianyungang city, which serves as the western starting point of the Eurasian Land Bridge and holds strategic significance for the implementation of the "Belt and Road" initiative (Feng, 2021). The scientific assessment and prediction of river water quality in Lianyungang city are crucial for sustainable water environment management in prosperous coastal zones adjacent to the Yellow Sea.

Scientific evaluation of water quality is crucial for ensuring the sustainable management of water resources in coastal cities, which exhibit unique characteristics and vulnerabilities, and confront multifaceted and comprehensive water pollution challenges. Traditionally, water quality assessment involved comparing values of various water parameters against standard levels prescribed for different grades, offering a singular description of the water environment's condition (Kothari et al., 2021). The Water Quality Index (WQI), first introduced in the 1960s, represents a comprehensive tool for assessing water quality. It transforms complex data from physical, chemical, and biological parameters into a single numerical value, providing a succinct representation of the overall condition of a water body (Lumb et al., 2011; Rana & Ganguly, 2020). Since its inception, the WQI approach has gained widespread acceptance and has led to the development of various models tailored for different types of water bodies, including lakes, rivers, groundwater, and wastewater (Mishra et al.,

2023; Uddin et al., 2022). Notable models include the United States National Sanitation Foundation Water Quality Index (NSFWQI), the British Columbia Water Quality Index (BCWQI), and the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI), all of which have received international recognition and application (Banda & Kumarasamy, 2020; Menberu et al., 2021; Talukdar et al., 2023). While these models are based on diverse parameters and assessment methodologies, they share the common aim of providing a quantitative and easily interpretable assessment of water quality status. However, given the frequent occurrence of water pollution incidents, there is an urgent need for rapid assessments of water quality in emergency situations, sometimes with only a limited number of parameters available (Gao et al., 2023; Yongo et al., 2023). Consequently, research efforts have focused on enhancing the efficiency of assessments and reduce analytical costs, by which the water quality assessment process were optimized by minimizing the number of parameters used in WQI models, achieved by selecting key parameters that reduce redundant information and measurement costs (Mu et al., 2024a, b; Sang et al., 2024). The application of key water parameters to evaluate water quality status is particularly suitable for resource-constrained environments and supports timely water quality management decisions (Zhao et al., 2024).

Multivariate statistical techniques are widely employed in water quality analysis. For example, Principal Component Analysis (PCA) is used to reduce data dimensionality by extracting principal variables and preserving as much information as possible, which aids in identifying dominant pollution sources (Mahanty et al., 2023). Factor Analysis (FA) helps simplify data and identify structures by focusing on the covariance among variables (Nsabimana & Li, 2023). However, these statistical methods typically assume linearity and normality in the data, which may not hold true for real-world datasets that can exhibit non-linearity and heteroscedasticity, thereby limiting the applicability of these traditional methods (Kamel & Abonazel, 2023). With advancements in computational power, machine learning models have become increasingly prominent in water quality prediction and analysis, due to their ability to handle large datasets characterized by non-linearity and high dimensionality (Aldrees et al., 2023; Haggerty et al., 2023). Notable examples include Random Forest (RF), which improves prediction accuracy by constructing multiple decision trees and outputting the average results (Sakaa et al., 2022); Support Vector Machine (SVM), which addresses complex and high-dimensional classification and regression problems by identifying the maximum margin between data categories (Koranga et al., 2021); and Artificial Neural Networks (ANNs), which mimic the human brain's processing approach by learning complex patterns between inputs and outputs to facilitate prediction (Mohammadpour et al., 2024). These machine learning techniques offer powerful alternatives to traditional statistical methods, which could contribute to the precise prediction of WQI. However, in some urgent situations, for example, sudden water pollution incidents, the most reliable models were essential to provide a quick assessment of water environment based on limit data, which should be selected among quantity statistical or machine learning models in advance.

Under the combined influence of natural processes and anthropogenic factors, water quality in rivers experiences seasonal fluctuations (Mondal et al., 2016; Xu et al., 2019). Climatic variables such as precipitation levels, temperature, and radiation exhibit seasonal variations that directly impact the physical and chemical properties of water bodies (Woolway et al., 2020). For instance, rising temperatures can accelerate biochemical reactions and foster algal growth, consequently affecting water quality (Xia et al., 2016). Seasonal patterns of rainfall influence the transportation of various pollutants from surface runoff into water bodies (Ding et al., 2022; Valtanen et al., 2015). Additionally, agricultural practices, including the application of fertilizers and pesticides, are major contributors to the eutrophication of water bodies (Khan & Mohammad, 2014; Liu et al., 2021). During rainy seasons, agricultural runoff may lead to a significant increase in nutrient concentrations, such as nitrogen and phosphorus, exacerbating the deterioration of water quality. Seasonal agricultural activities, such as planting and harvesting, also exert a significant influence on water quality (Sánchez-Cortés & Chavero, 2011). Therefore, the key water parameters influencing river environment varied in different seasons, which should be comprehensively considered in order to develop more effective water resource management strategies. However, to our knowledge, the

selection of key water parameters in previous studies rarely considered these season variations instead they assumed influenced water parameter groups to be identical, which may lead to less precision of WQI prediction or need more input variables.

In this study, we collected a 4-year water quality dataset (2020–2023) comprising nine water quality indicators from 14 monitoring stations in Lianyungang city, Jiangsu Province, a coastal city adjacent to the Yellow Sea. The main objectives of our study were to (1) develop WQI prediction models based on various input variable groups across different seasons, (2) identify the key water parameters and select the most reliable models for seasonal WQI prediction, and (3) assess the spatiotemporal variations of key water parameters by the Seasonal-Trend decomposition procedure based on Loess (STL). Our research is expected to contribute to effective and efficient river management in coastal cities adjacent to the Yellow Sea. Additionally, the methodological framework developed is also applicable to other rivers worldwide.

Material and methods

Study area

Lianyungang city (118° 35′ to 119° 33′ E, 33° 58′ to 35° 07′ N) (Fig. 1) is a major coastal city located in the eastern region of Jiangsu Province, China (Shi et al., 2012). Situated on the northern edge of the Yangtze River Delta, with the Yellow Sea to its east and the central Jiangsu Plain to its south, Lianyungang is endowed with abundant fishery and marine resources (Liu et al., 2023). Covering an area of approximately 7614 km^2, the city has a total population of around 4.6 million and boasts a coastline that stretches over 211 km (Sun et al., 2012). The topography features a noticeable west-to-east slope. The western and northern edges of the city extend into the undulating terrain of the Subei Hills and Yishan Mountains, where elevations are relatively higher. In contrast, the eastern and southern regions consist predominantly of flat plains, which serve as the principal agricultural production zones.

The climate of Lianyungang is characterized by a northern subtropical monsoon pattern, marking a transition between temperate and subtropical conditions with distinct monsoon influences (Zhang et al., 2024b). The summer months are dominated by the southeast monsoon, which brings humid and rainy weather (Chen et al., 2021). In contrast, the winter season is cooler and drier, influenced by the northwest monsoon. The city's annual average temperature typically ranges from approximately 14.0 to 15.5 °C (Fang et al., 2017). July and August are the warmest months, with average temperatures reaching around 27 °C. January is the coldest month, with average temperatures ranging from 0 to − 1 °C. Lianyungang receives an average annual precipitation of 800 to 1000 mm (Lin et al., 2018). The majority of this rainfall occurs during the summer months, particularly from June to August, when typhoons may bring intense downpours (Yuan et al., 2023).

Crop cultivation is a major component of the agricultural landscape in Lianyungang (Yang et al., 2015). The city primarily produces cereal grains such as wheat, rice, and corn, along with cash crops like cotton and peanuts (Liu et al., 2022). Fruits, including apples and citrus, are also significant agricultural products (Alim et al., 2023). The coastal stretches of Lianyungang are characterized by vast tidal flats and saline-alkali lands, which are extensively used for salt harvesting and aquaculture farming (Wei et al., 2017; Zhu et al., 2016). The soils near the coastlines and rivers are predominantly sandy loam, known for their good drainage capabilities but limited water retention capacity, making them particularly suitable for vegetable cultivation (Mao et al., 2017).

Water quality dataset and WQI calculation

The water quality dataset collected in the study consisted 14 monitoring stations in Lianyungang city (Table S1), which were collected from Environmental Protection Department in Jiangsu province, including nine water quality parameters, accordingly, Water Temperature (WT), pH, Dissolved Oxygen (DO), Electrical Conductivity (EC), Turbidity (Tur), Permanganate Index (COM), Ammonia Nitrogen (AN), Total Phosphorus (TP), and Total Nitrogen (TN). The dataset ranged from 2020 November to 2023 August and included 12,821 records in total. The WQI values were calculated by the following:

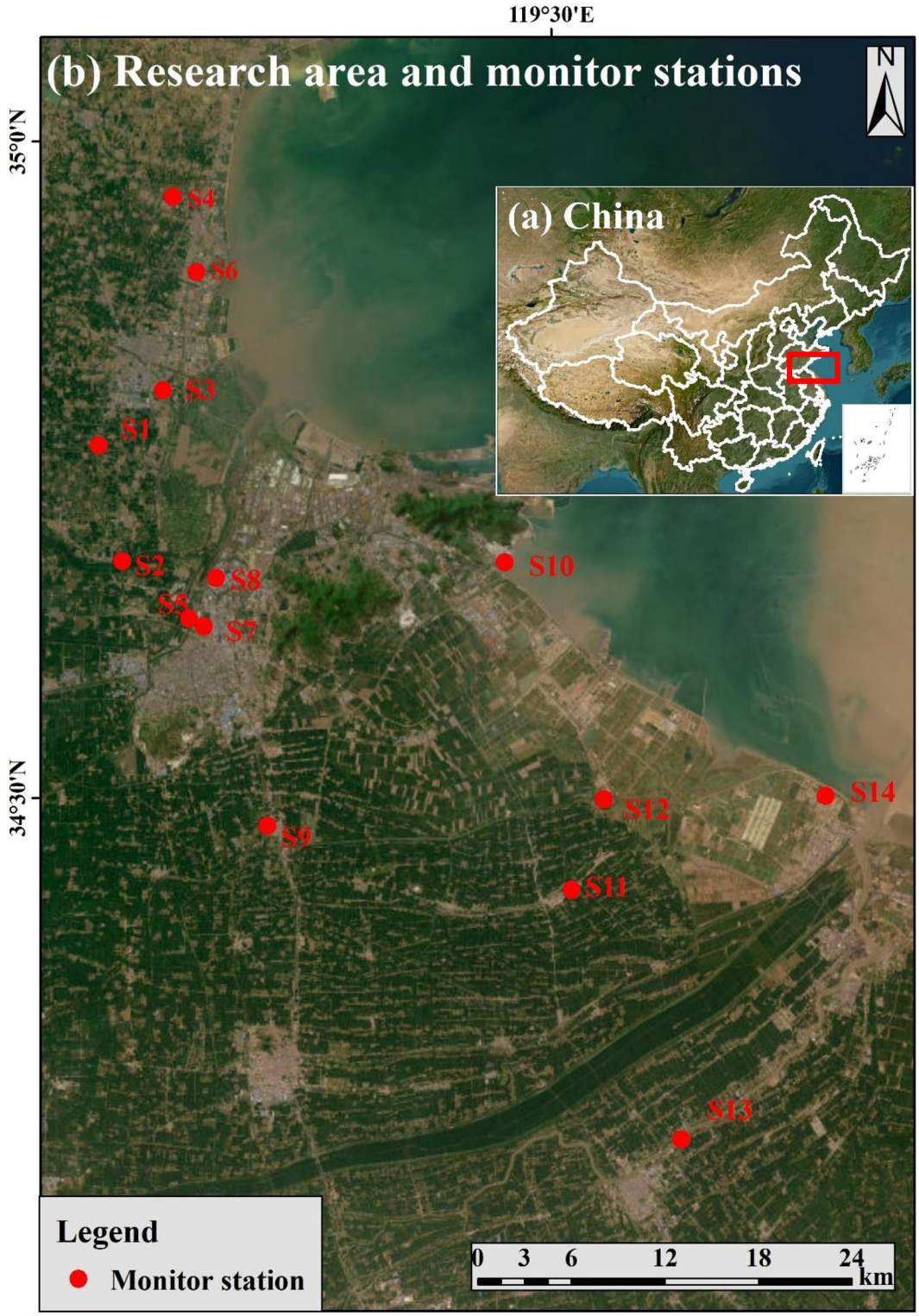


Fig. 1 The map of research area and monitor stations in the study

$$WQI = \frac{\sum_{i=1}^{n} C_i P_i}{\sum_{i=1}^{n} P_i} \quad (1)$$

where the C_i represents the normalized value of parameter i and P_i represents the weight of parameter i. The normalized value and weight of all water quality parameters are shown in Table S2, which were recommended by previous research (Sun et al., 2016; Wu et al., 2018). Based on the calculated WQI values, the water quality could be separated into five levels (Table S3), accordingly, "excellent" level (91–100), "good" level (71–90), "medium" level (51–70), "low" level (26–50), and "bad" level (0–25).

Model development and evaluation

To determine the key water parameters and the most reliable model for efficiently predicting the WQI values in different seasons, the dataset was divided according to the four seasons: March to May for the spring dataset, June to August for the summer dataset, September to November for the autumn dataset, and December to February in next year for the winter dataset. The different groups of input variables, including partial or entire water quality datasets, were determined using Pearson's correlation analysis, in which the corresponding water quality parameters were sequentially added to the input groups based on their correlation coefficients in descending order. The WQI prediction models were developed using both statistical methods and machine learning algorithms, including Linear Model (LM), Generalized Linear Model (GLM), Generalized Additive Model (GAM), Artificial Neural Networks (ANNs), Self-Organizing Map (SOM), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), Extreme Gradient Boosting (XGB), and Stochastic Gradient Boosting (SGB).

The LM is a fundamental statistical model that describes the linear relationship between independent variables and a dependent variable using the least squares method for rapid parameter estimation (Ding, 2023). However, it has limitations, particularly its inability to account for non-linear relationships between variables. The GLM expands upon the linear model by allowing the dependent variable's distribution to belong to the exponential family, such as binomial or Poisson distributions (Dobson & Barnett, 2018). It also uses a link function to connect the expected value of the dependent variable to a linear combination of independent variables, enhancing its flexibility. Building further on the GLM, the GAM allows for the inclusion of independent variables in non-parametric or semi-parametric forms. This model combines the interpretability of linear models with the flexibility of non-linear models, maintaining a degree of interpretability while accommodating more complex relationships (Beck & Jackman, 1998).

The ANNs emulate the structure and functioning of biological neural networks to process information. It includes an input layer, one or more hidden layers, and an output layer, each consisting of multiple neurons interconnected through weighted connections. This configuration endows the ANNs with powerful data fitting and feature learning capabilities (Dongare et al., 2012). The SOM is a dimensionality reduction algorithm that excels at mapping complex, high-dimensional data onto typically two-dimensional topological structures based on learning data from the input space. It uses a competitive learning strategy, relying on the competition between neurons to gradually optimize the neural network, and uses a neighbor relation function to maintain the topological structure of the input space. The training process of SOM does not depend on specific data distributions, which endows it with robust generalization capabilities (Laaksonen et al., 2004). The KNN algorithm classifies new sample points based on their proximity, as measured by a distance metric, to the majority class or average value of the K nearest training samples, which can be employed for both classification and regression tasks (Zhang, 2022). The SVM primarily tackles problems by identifying an optimal hyperplane within the dataset that maximizes the margin between distinct classes of data. With careful selection of kernel functions and regularization parameters, SVMs can effectively handle high-dimensional data and non-linear problems (Ghosh et al., 2019).

The RF is an ensemble technique that constructs multiple decision trees and combines their predictions through averaging or majority voting to enhance prediction accuracy and stability. By integrating multiple decision trees, the model achieves higher accuracy and is less susceptible to overfitting compared to a single decision tree (Shaikhina et al., 2019). The XGB is an implementation of boosted trees where it sequentially incorporates trees, each new tree aiming

to rectify the prediction errors of all preceding trees. It also incorporates regularization terms to control model complexity and prevent overfitting (Zhao et al., 2020). The SGB is an ensemble method that leverages a gradient boosting mechanism, progressively adding decision trees to correct the residuals of previous trees. The SGB differs in that it employs a randomly subsampled subset of data when constructing each tree, introducing randomness that enhances model generalization and flexibility for various problems, such as regression and classification (Friedman, 2002). These three models diverge in their ensemble strategies: RF employs bagging, while XGB and SGB utilize boosting (Sahin, 2020).

The model preferences were comprehensively evaluated using Mean Absolute Error (MAE), Nash–Sutcliffe Efficiency Coefficient (NSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). The MAE and RMSE provide an intuitive understanding of the extent of the errors. The NSE is a widely accepted method in hydrology and environmental modeling for assessing the accuracy of model predictions. It provides a normalized score ranging from 0 to 1, reflecting the quality of the model's performance, where a score of 1 indicates perfect prediction. The R^2 offers a proportional explanation of the model's predictive ability and can be interpreted as the percentage of the variation in the data that the model can explain.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|P_i - A_i\right| \tag{2}$$

$$NSE = 1 - \frac{\sum_{i=1}^{n}\left(A_i - P_i\right)^2}{\sum_{i=1}^{n}\left(A_i - \overline{A_i}\right)^2} \tag{3}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(P_i - A_i\right)^2}{n}} \tag{4}$$

$$R^2 = \frac{\left[\sum_{i=1}^{n}\left(A_i - \overline{A_i}\right)\left(P_i - \overline{P_i}\right)\right]^2}{\sum_{i=1}^{n}\left(A_i - \overline{A_i}\right)^2\sum_{i=1}^{n}\left(P_i - \overline{P_i}\right)^2} \tag{5}$$

where A_i and P_i represent actual and predicted WQI at the ith data; $\overline{A_i}$ and $\overline{P_i}$ represent the average value of actual and predicted WQI, respectively; n represents the number of WQI records. The methodological framework of the study is shown in Fig. 2. ALL machine learning models in WQI prediction were developed by the "caret" package (Kuhn, 2008) and all data processes were completed in the R statistical computing platform.

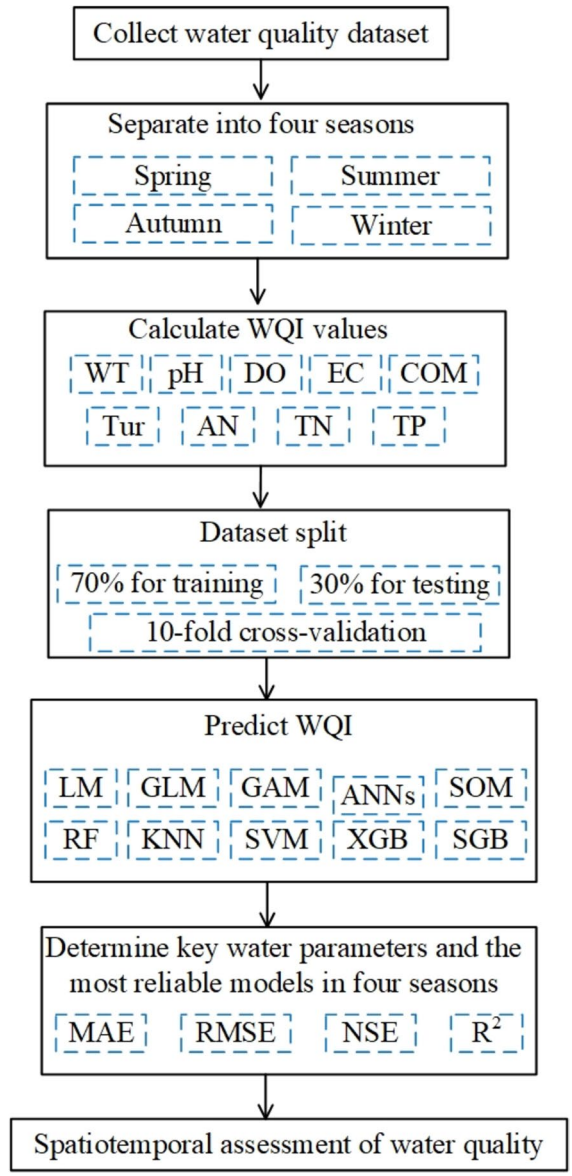


Fig. 2 The methodological framework for WQI prediction and key water quality parameters selection

Variations analysis of key water parameters

To assess variations of water quality in the city, the STL method was applied to decompose the dataset of key water parameters into trend and seasonal components. The STL method consists of inner and outer loops, based on a non-parametric iterative procedure using local weighted regression (LOESS) (Cleveland et al., 1988). It extracts long-term trend, high-frequency seasonal, and random residual components from daily time series. In the inner loop, long-term trend and seasonal components are extracted through consecutive locally weighted curve fitting. The outer loop employs robust estimation, where outliers deviating more from normal values are given less weight in the estimation, ensuring that long-term trend and seasonal components are minimally affected by outliers. The iterative process stops when all components are extracted and convergence is verified. This approach allows for a comprehensive analysis of the temporal and spatial distribution characteristics of key water parameters, providing valuable insights into water quality changes over time. Accordingly, the STL method included the following six steps: detrending of the time series, cycle-subseries smoothing, low-pass filtering of smoothed cycle-subseries, detrending of smoothed cycle-subseries, deseasonalizing the series, and trend smoothing of the extracted trend component.

Based on the STL methods, water quality dataset could be divided as follows:

$$C = C_T + C_S + C_R \tag{6}$$

Here, C represents the actual value of the water quality indicator (mg/L); C_T is the low-frequency trend component decomposed from the original data (mg/L); C_S is the high-frequency seasonal component decomposed from the original data (mg/L); C_R refers to the residual component of the water quality observations that does not conform to long-term trend and seasonal characteristics (mg/L). Generally, larger residual values correspond to anomalies in the water quality data.

Results and discussion

Summary of water quality dataset

The dataset for analysis is robust, with the number of records ranging from 761 at site S5 to 982 at site S3 (Table 1). The WT shows relative consistency across the sites, fluctuating within a range of 16.7 to 18.5 °C. The close proximity of the monitoring locations contributes to the lack of extreme temperature variations in the monitored water bodies. The pH values range from 7.51 to 8.16, with site S10 recording the highest value of 8.16, indicative of alkaline water quality. The DO content shows significant variability, with site S10 exhibiting the highest level at 11.29 mg/L and site S13 the lowest at 6.27 mg/L. This variation may be attributed to differences in biological activity and levels of organic pollution across the sites. The EC values vary substantially, ranging from 591 µS/cm at site S13 to 2722 µS/cm at site S10. The elevated conductivity at S10 could indicate a higher degree of mineralization or salinity in the water.

The Tur levels also show significant discrepancies, with site S8 registering the lowest value at 15.81 NTU and site S14 the highest at 58.4 NTU. Elevated Tur concentration may result from the presence of suspended particles, such as silt or organic matter, in the water. Site S2 exhibits the lowest COD at 4.24 mg/L, while site S10 has the highest at 6.16 mg/L, reflecting differences in organic pollution levels. The concentrations of AN and TP vary across sites, with site S8 having the highest AN level at 0.55 mg/L and site S13 having the highest TP at 0.19 mg/L. These variations may be linked to agricultural activities or industrial discharges in the vicinity of each site. The TN content ranges from 2.04 mg/L at S13 to 3.81 mg/L at S1, reflecting the differences in nitrogen sources across the locations.

WQI characteristics

WQI values and grades in the city

The WQI revealed that site S13 has the lowest score of 49.9, indicating degraded water quality conditions, whereas site S2 has the highest score of 68.7, reflecting comparatively better water quality. Table 2 illustrates the distribution of WQI grades for different seasons, including Excellent, Good, Medium, Low, and

Table 1 Summary of nine water quality parameters and WQI at each monitoring station

Code	WT (℃)	pH	DO (mg/L)	EC (μS/cm)	Tur (NTU)	COM (mg/L)	AN (mg/L)	TP (mg/L)	TN (mg/L)	WQI	Records number
S1	17.2 ± 8.7	7.51 ± 0.31	8.78 ± 4.67	959 ± 285	34.0 ± 28.0	5.50 ± 1.29	0.44 ± 0.71	0.15 ± 0.08	3.98 ± 1.31	59.4 ± 11.3	980
S2	16.9 ± 8.6	8.01 ± 0.43	8.62 ± 3.61	640 ± 107	39.3 ± 35.5	4.24 ± 0.85	0.04 ± 0.03	0.11 ± 0.08	3.08 ± 1.57	68.0 ± 7.4	883
S3	17.1 ± 8.6	7.78 ± 0.49	9.69 ± 3.80	696 ± 118	25.8 ± 16.2	5.01 ± 1.22	0.30 ± 0.46	0.13 ± 0.10	3.72 ± 1.15	64.1 ± 10.4	982
S4	16.7 ± 8.7	7.98 ± 0.61	9.56 ± 3.54	705 ± 131	21.6 ± 16.0	4.54 ± 0.89	0.09 ± 0.13	0.13 ± 0.09	3.85 ± 1.30	67.9 ± 7.7	893
S5	18.4 ± 8.7	7.67 ± 0.42	7.29 ± 4.31	786 ± 197	22.0 ± 25.2	5.17 ± 1.33	0.28 ± 0.40	0.18 ± 0.08	3.10 ± 1.57	61.8 ± 12.6	761
S6	17.3 ± 8.6	7.91 ± 0.53	10.47 ± 4.73	661 ± 192	26.6 ± 26.7	5.01 ± 1.29	0.26 ± 0.34	0.11 ± 0.08	2.55 ± 1.04	65.4 ± 10.7	823
S7	18.4 ± 8.6	8.07 ± 0.55	8.84 ± 4.50	722 ± 138	15.9 ± 12.6	5.20 ± 1.58	0.19 ± 0.33	0.12 ± 0.08	2.48 ± 1.14	66.4 ± 12.7	870
S8	18.0 ± 8.0	7.53 ± 0.41	9.20 ± 3.54	1374 ± 465	15.8 ± 15.8	5.46 ± 1.23	0.55 ± 0.87	0.14 ± 0.10	6.54 ± 2.23	62.7 ± 11.2	951
S9	17.3 ± 8.6	7.88 ± 0.53	9.45 ± 4.98	1845 ± 258	19.3 ± 16.0	5.57 ± 1.40	0.54 ± 0.76	0.16 ± 0.11	3.73 ± 1.67	60.7 ± 13.5	899
S10	16.8 ± 8.8	8.16 ± 0.41	11.29 ± 4.20	2722 ± 1170	24.0 ± 35.0	6.16 ± 1.40	0.32 ± 0.52	0.08 ± 0.05	4.10 ± 1.33	62.6 ± 7.7	937
S11	17.3 ± 8.7	7.83 ± 0.42	7.62 ± 3.82	1040 ± 211	34.0 ± 22.5	5.21 ± 1.46	0.22 ± 0.33	0.14 ± 0.09	3.04 ± 1.27	60.3 ± 11.5	981
S12	17.6 ± 9.1	7.93 ± 0.44	8.29 ± 3.97	1026 ± 282	25.5 ± 22.7	4.92 ± 1.25	0.21 ± 0.35	0.14 ± 0.18	2.90 ± 1.43	63.3 ± 11.5	959
S13	17.6 ± 8.5	7.79 ± 0.41	6.27 ± 2.85	915 ± 267	294 ± 121	5.89 ± 1.33	0.36 ± 0.32	0.19 ± 0.08	3.04 ± 0.77	49.9 ± 8.9	934
S14	16.9 ± 8.3	7.95 ± 0.41	8.13 ± 3.32	2141 ± 1245	58.4 ± 38.1	5.67 ± 1.15	0.21 ± 0.36	0.12 ± 0.09	3.60 ± 0.95	58.4 ± 9.9	968

Table 2 The percentages of WQI grades at all monitoring stations in four seasons

Station	Spring					Summer					Autumn					Winter				
	E	G	M	L	B	E	G	M	L	B	E	G	M	L	B	E	G	M	L	B
S1	0	36.7	61.0	2.3	0	0	7.6	32.8	58.4	1.2	0	6.5	72.4	21.1	0	0	6.8	90.6	2.6	0
S2	0	50.6	48.7	0.7	0	0	14.5	77.1	8.4	0	0	29.6	70.4	0	0	0	69.8	30.2	0	0
S3	0	44.7	52.6	2.6	0	0	9.4	51.8	38.4	0.4	0	36.6	58.5	4.9	0	0	30.2	69.4	0.4	0
S4	0	64.0	36.0	0	0	0	11.5	67.0	21.5	0	0	45.6	53.4	1.0	0	0	60.9	39.1	0	0
S5	0	55.2	44.2	0.6	0	0	12.6	40.6	44.3	2.5	0	19.6	69.8	10.6	0	0	11.9	88.1	0	0
S6	0	65.0	33.9	1.1	0	0	17.3	50.4	32.3	0	0	42.6	51.5	5.9	0	0	7.9	92.1	0	0
S7	0	80.1	19.9	0	0	0	27.4	29.9	41.9	0.8	0	28.6	64.6	6.8	0	0	57.4	42.6	0	0
S8	0	37.2	59.0	3.8	0	0	5.6	57.0	36.1	1.3	0	33.0	53.0	14.0	0	0	22.4	75.7	1.9	0
S9	0	43.7	51.0	5.3	0	0	7.1	32.2	59.4	1.3	0	23.5	56.9	19.6	0	0	19.6	75.4	5.0	0
S10	0	16.8	78.9	3.9	0.4	0	2.1	83.5	13.9	0.5	0	9.1	87.2	3.7	0	0	6.1	86.2	7.7	0
S11	0	34.7	64.6	0.7	0	0	0.4	40.1	55.0	4.5	0	5.4	84.4	10.2	0	0	5.4	94.6	0	0
S12	0	45.1	54.9	0	0	0	13.3	39.4	46.1	1.2	0	10.4	86.6	3.0	0	0	34.5	65.5	0	0
S13	0	0	70.5	29.5	0	0	0	12.6	74.8	12.6	0	0	66.1	33.9	0	0	0.4	64.3	35.3	0
S14	0	4.1	95.9	0	0	0	0	52.9	39.9	7.2	0	1.0	92.0	7.0	0	0	0	99.6	0.4	0

The letters E, G, M, L, and B represent Excellent, Good, Medium, Low, and Bad water quality, respectively, which were the five levels of the WQI used in the research. All the numerical values in the table indicate the percentage of records falling within the corresponding WQI classification

Bad grades, showing substantial seasonal fluctuations in water quality status across monitoring stations.

During spring, the majority of the sites were classified as Medium and Good levels, with no instances of Excellent grades. In summer, the percentage of sites rated as Medium and Good was lower than in spring, with most stations recording approximately half of the observations in these two grades. However, there was a significant increase in the proportion of Low and Bad grades during this period, which indicated the deterioration of water quality in the season. Water quality conditions improved in autumn compared to summer, with numerous sites such as S2 and S12 showing elevated percentages in the Good and Medium grades. A noticeable improvement in water quality occurred during winter, as many sites experienced a substantial increase in the proportions of Good and Medium grades. Compared to other seasons, the summer period generally exhibited inferior water quality, with a higher prevalence of Low grades. In contrast, the spring and winter seasons tended to show enhanced water quality, with the majority of sites concentrated in the Medium and Good categories. The autumn season demonstrated mixed water quality outcomes but was overall superior to the summer period.

Correlation between water quality parameters and WQI

In order to determine different combinations of key water parameters which had important effects on WQI prediction in four seasons, the correlation between different water quality parameters and WQI values is analyzed (Fig. 3). The different groups of input variables to the WQI prediction models in four seasons were determined by descending order of the correlation coefficients between water parameters and WQI values, for example, Tur was the most correlated to WQI (Cor = 0.57) in Spring; then, it was the selected water parameter when only used one variable to predict WQI values.

During the spring season (Table 3), the correlation coefficient between Tur and the WQI was 0.57 while correlation coefficients between AN and DO with WQI are closely aligned at 0.54 and 0.48, respectively. Similarly, in the summer (Table 4) and autumn (Table 5) months, the water quality parameters exhibiting the strongest correlation with WQI remain to be DO, AN, pH, and TP despite variations existing in correlation coefficients. In the winter (Table 6) period, however, the influence of DO on WQI weakened, with the Tur emerging as the

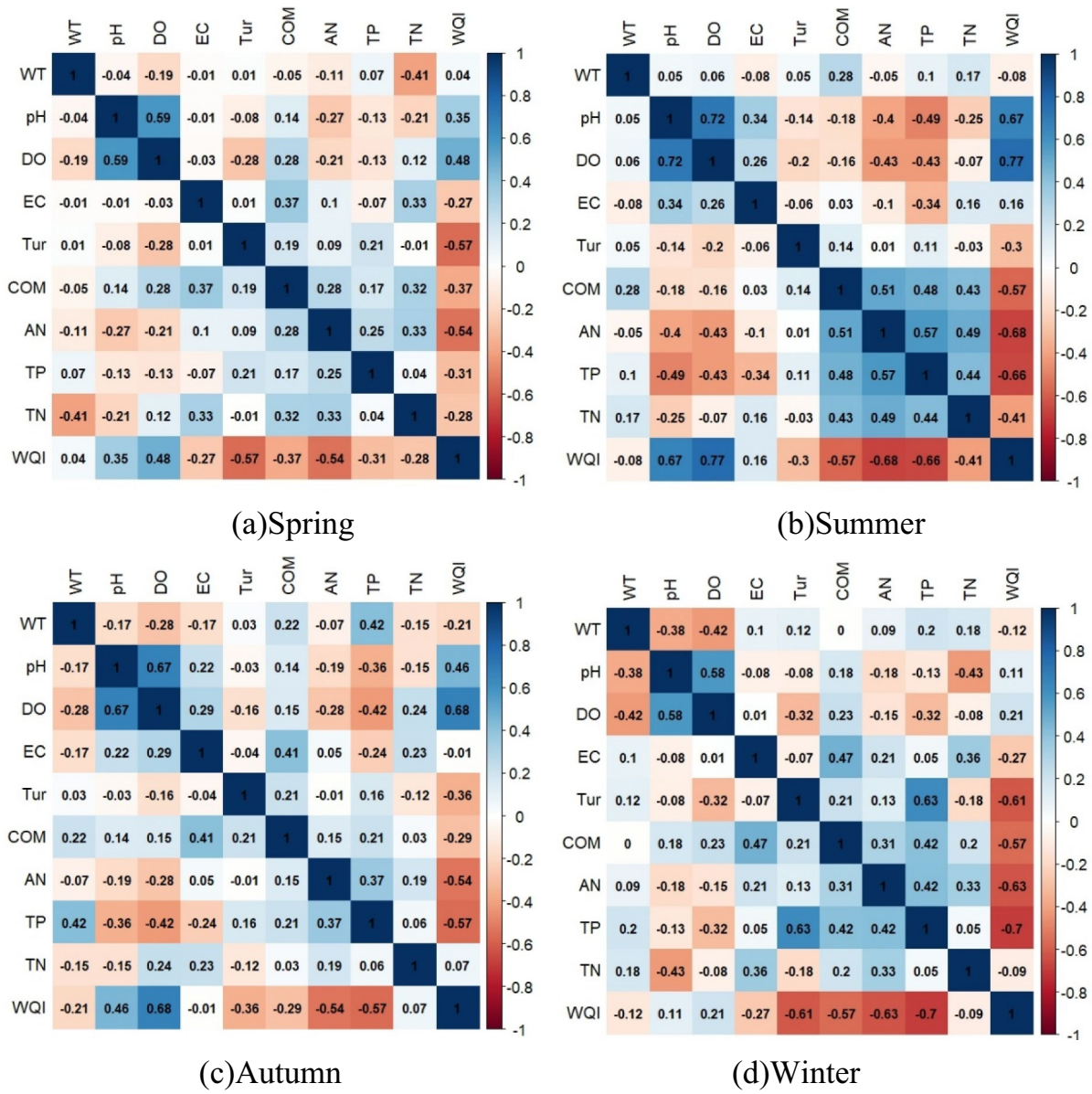


Fig. 3 Correlation coefficients between water parameters and WQI in spring, summer, autumn, and winter, respectively, **a** Spring, **b** Summer, **c** Autumn, **d** Winter

third most significant water quality factor affecting WQI. The TP remains a crucial indicator across all seasons (Table S4). The level of DO is significantly influenced by temperature, with higher temperatures leading to decreased DO levels but increased biological demand, which may be the cause of the important effects of DO on the WQI in seasons except the winter (Nong et al., 2020). AN and TP demonstrate a stronger correlation with WQI across all seasons, likely attributable to rapid biological growth during these seasons, which heightens the demand for nutrients (Kamrunnahar et al., 2017).

Table 3 Combinations of input variables to predict the WQI in spring

Group number	Input variables
1	Tur
2	Tur, AN
3	Tur, AN, DO
4	Tur, AN, DO, COM
5	Tur, AN, DO, COM, pH
6	Tur, AN, DO, COM, pH, TP
7	Tur, AN, DO, COM, pH, TP, TN
8	Tur, AN, DO, COM, pH, TP, TN, EC
9	Tur, AN, DO, COM, pH, TP, TN, EC, WT

Table 4 Combinations of input variables to predict the WQI in summer

Group number	Input variables
1	DO
2	DO, AN
3	DO, AN, pH
4	DO, AN, pH, TP
5	DO, AN, pH, TP, COM
6	DO, AN, pH, TP, COM, TN
7	DO, AN, pH, TP, COM, TN, Tur
8	DO, AN, pH, TP, COM, TN, Tur, EC
9	DO, AN, pH, TP, COM, TN, Tur, EC, WT

Table 5 Combinations of input variables to predict the WQI in autumn

Group number	Input variables
1	DO
2	DO, TP
3	DO, TP, AN
4	DO, TP, AN, pH
5	DO, TP, AN, pH, Tur
6	DO, TP, AN, pH, Tur, COM
7	DO, TP, AN, pH, Tur, COM, WT
8	DO, TP, AN, pH, Tur, COM, WT, TN
9	DO, TP, AN, pH, Tur, COM, WT, TN, EC

Table 6 Combinations of input variables to predict the WQI in winter

Group number	Input variables
1	TP
2	TP, AN
3	TP, AN, Tur
4	TP, AN, Tur, COM
5	TP, AN, Tur, COM, EC
6	TP, AN, Tur, COM, EC, DO
7	TP, AN, Tur, COM, EC, DO, WT
8	TP, AN, Tur, COM, EC, DO, WT, pH
9	TP, AN, Tur, COM, EC, DO, WT, pH, TN

Table 7 Details of tuning parameters and settings of the optimal models for WQI predicting

Models	Tuning parameters
LM	Intercept = True
GLM	No tuning parameters
GAM	Fitting select = TRUE, method = GCV.Cp
ANNs	Size = 5, decay = 1×10^{-4}
SOM	xdim = 3, ydim = 4, user.weights = 0.5, topo = hexagonal
KNN	kmax = 2, distance = 2, kernel = optimal
SVM	Radial basis function kernel, sigma = 0.153, C = 1
RF	mtry = 5
XGB	nrounds = 100, lambda = 1×10^{-4}, alpha = 0.1, eta = 0.3
SGB	n.trees = 150, interaction.depth = 3, shrinkage = 0.1, n.minobsinnode = 10

All optimal models were selected by the smallest RMSE except the LM

Model evaluation and uncertainty analysis

Comparisons of model performances

All models were validated by tenfold cross-validation, and the optimized tuning parameters are summarized in Table 7. For the spring dataset, the performance of the LM model shows a gradual improvement as the number of input variables increases from one to nine (Table S5). With just one variable, the metrics are MAE = 0.076, NSE = 0.325, RMSE = 0.102, and $R^2 = 0.325$ in the test phase. When expanded to nine variables, the performance metrics improve to MAE = 0.050,

NSE = 0.721, RMSE = 0.066, and $R^2 = 0.721$. As more variables are incorporated, the model's explanatory power shows a limited increase. In contrast, the GAM model demonstrates relatively superior performance, particularly notable in its R^2, which increases to 0.985 with nine variables. With the same number of input variables, the ANN model (forward-feeding back-propagation neural network) achieves an MAE of 0.012 and an R^2 of 0.985, demonstrating high prediction accuracy and model quality. Both the SVM and RF models exhibit formidable performance, with the RF model achieving an impressive MAE of 0.006 and an R^2 of 0.995 when configured with nine input variables. This underscores its suitability for handling complex and high-dimensional datasets.

The tree-based boosting methods, specifically XGB and SGB, approached perfection with nine input variables, highlighting their remarkable efficiency in model accuracy and explanatory power. Conversely, the SOM and KNN exhibited inconsistent performance under certain configurations. As the number of variables increased, the performance of the KNN model showed noticeable improvement. This inconsistency underscores the possible instability or sensitivity of these instance-based learning approaches to the configuration of input variables, emphasizing the importance of careful evaluation and tuning for optimal performance in specific data contexts (Wani et al., 2017). In general, as the number of input variables increases, the majority of the models demonstrate performance enhancements. This trend is particularly evident in models such as RF, XGB, SGB, and ANNs, which all exhibit significant improvements in their predictive abilities.

The analysis of the summer, autumn, and winter datasets revealed similar performance patterns and characteristics across the different models when compared to the spring dataset. Notably, the models generally exhibited superior performance on the winter data, with XGB and SGB standing out as exceptional performers, achieving outstanding results across almost all metrics. In contrast, when evaluated on the annual dataset, which encompasses data from all seasons, the models' performance tended to be relatively lower compared to their performance on the season-specific datasets. This decrease in performance could be attributed to the increased seasonal variations present within the annual data, which introduces more complexity and variability for the models to handle effectively.

Selection of key water parameters and the most reliable models

Selecting an appropriate model architecture and determining the optimal number of input variables are crucial for enhancing prediction accuracy, as these factors directly affect model performance. The key water parameters for the models were chosen to achieve an R^2 of 0.9 with the fewest variables possible. The most reliable models were identified based on a combination of minimal key water parameters and maximal prediction precision. Table 8 summarizes the key water parameters and the most reliable models across four seasons, with the number of key water parameters ranging from two to three in all seasons. This approach underscores the cost-effectiveness of considering seasonal variations in predicting the WQI.

Figure 4 presents a series of scatter plots comparing actual and predicted WQI values, derived from various machine learning models across different seasonal periods. In each plot, a red line represents the ideal scenario of perfect prediction, where predicted values exactly match the actual values. During the spring season, the data points from the RF and XGB models, each using three input variables, are closely aligned with the red line. This indicates that these models are particularly reliable and deliver high prediction accuracy in the spring. In the summer, the points representing the KNN and RF models are more tightly clustered around the red line, reflecting superior prediction performance. Conversely, the SGB and XGB models show a greater dispersion of points, indicating slightly poorer performance during this season. In the autumn, the RF model's points are

Table 8 The key water parameters determined by machine learning models for different seasons and the whole dataset ($R^2 > 0.9$)

Dataset	Key water parameters	Models
Spring	Tur, AN, DO	RF, XGB, SGB
Summer	DO, AN	RF, XGB, KNN, SGB
Autumn	DO, TP, AN	RF
Winter	TP, AN, Tur	SVM, RF, XGB, KNN, SGB, GAM

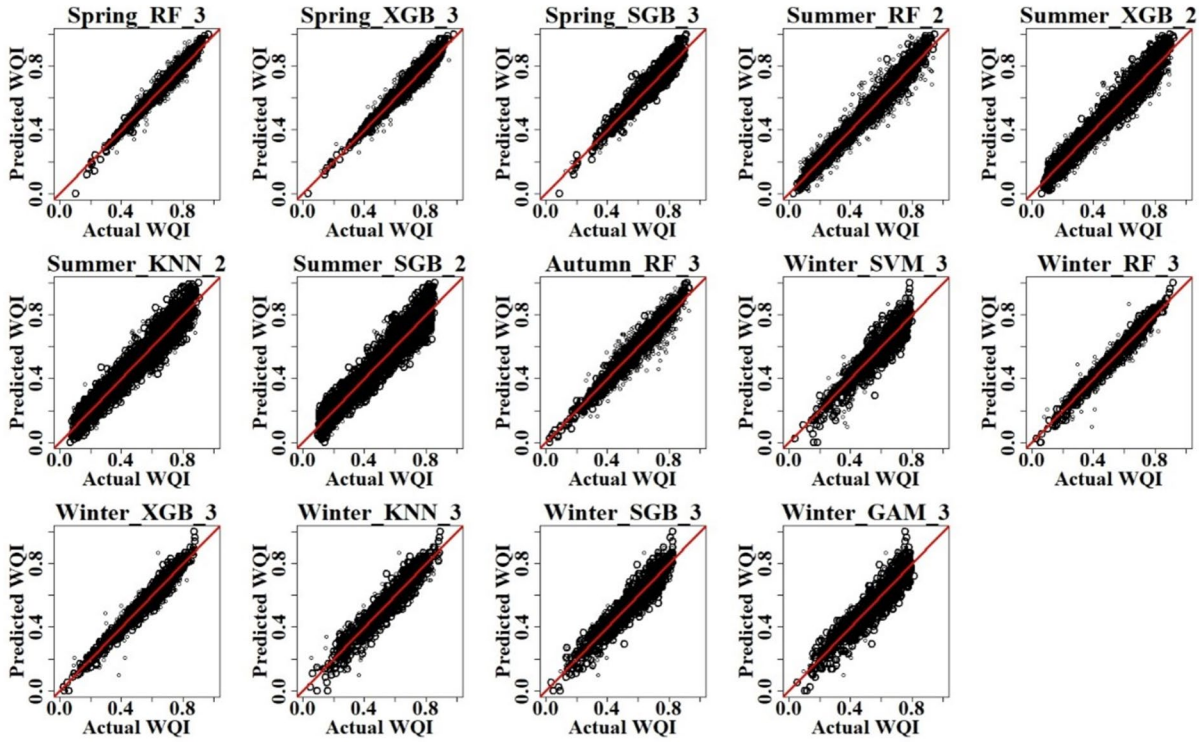


Fig. 4 The actual vs. predicted WQI which were predicted by key water parameters in four seasons

nearest to the red line, demonstrating exemplary predictive accuracy. During the winter season, the points for the RF and XGB models are closely grouped near the red line, suggesting good prediction accuracy. However, the points for the KNN and SGB models are more scattered, implying a reduction in predictive performance. Different models exhibit noticeable variations in performance across different seasons, with the RF and XGB models demonstrating relatively stable and superior performance across multiple seasons. Furthermore, the performance of each model is influenced by seasonal changes, which may be related to the varying characteristics of water quality during different seasons. Through comparative analysis, it becomes evident that selecting models tailored to specific seasons is crucial for enhancing prediction accuracy.

Model performance in classifying WQI grades

Due to the absence of water quality records assessed as Excellent grade, the performance of the most reliable model was evaluated across four categories: Good, Medium, Low, and Bad (Fig. 5). For the Good grade, the models generally demonstrated high predictive accuracy during the spring and winter, with most models achieving accuracy rates exceeding 80%. Although a slight decline in accuracy was observed during the summer and autumn, the majority of models maintained a respectable accuracy level above 70%. The Medium grade was consistently misjudged across all four seasons. However, the models exhibited outstanding performance in predicting the Medium grades, with nearly all data points achieving accuracies approaching or reaching 100%. This remarkable performance suggests that the models have effectively learned the features associated with the Medium grades, enabling accurate identification of water quality conditions within this category. When predicting the Low grades during the spring season, the models generally showed accuracies above 80%, except for the SGB model in spring and the SVM in winter. The highest prediction accuracy was noted in summer, nearly approaching 100%. Due to the absence of records assessed as Bad water quality in autumn and winter, the prediction performance

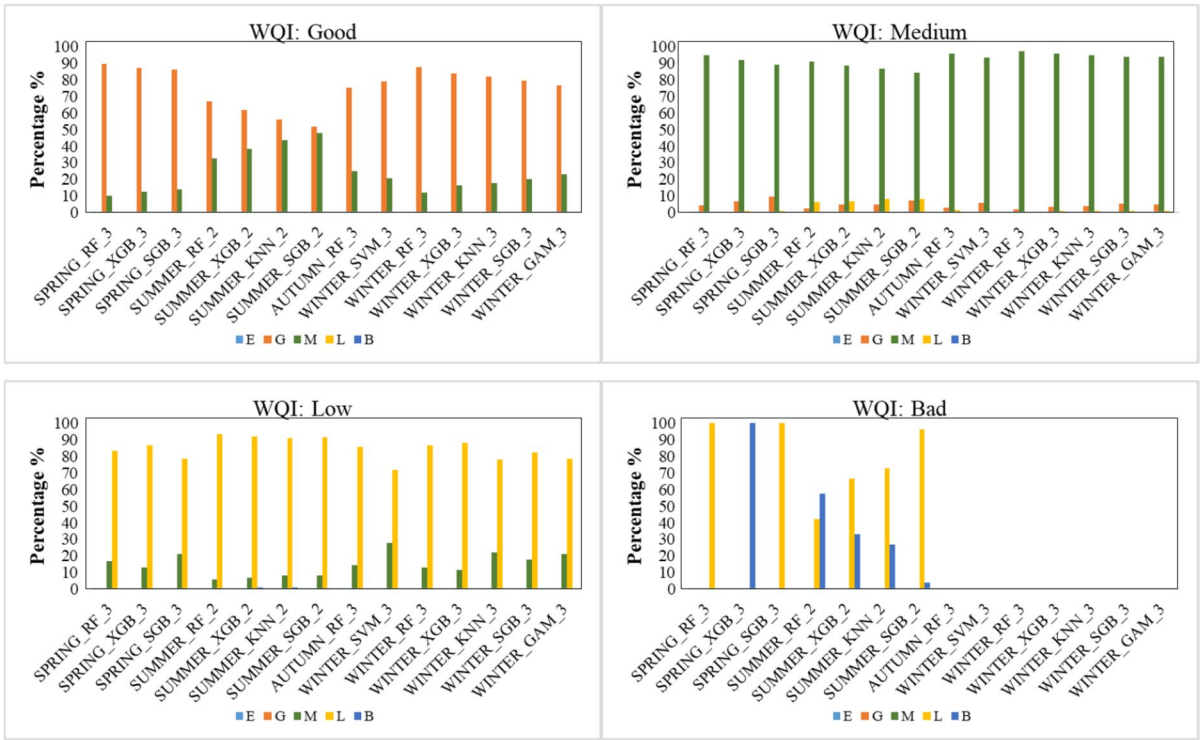


Fig. 5 The performance of WQI classification based on key water parameters determined by the most reliable models in four seasons

for this grade was analyzed only in spring and summer. Unlike other water quality grades, the models exhibited the lowest accuracy in predicting Bad grades, typically below 50%. This posed a challenge for the models to consistently identify the features associated with extremely poor water quality conditions. The majority of Bad grades were misjudged as Low grades, except for predictions from the XGB model in spring, which achieved 100% accuracy.

The performance of the models is significantly affected by seasonal variations. For instance, the models display enhanced capability in predicting low water quality during the summer, while their accuracy improves when predicting good water quality in the spring and winter. These variations can be attributed to the distinct features associated with seasonal changes in water quality. For example, during the summer months, higher temperatures and increased biological activity can lead to a greater accumulation of pollutants (Li et al., 2024). The impact of seasonal variations on water quality, alongside the adaptability of the models, highlights the inherent complexity of water quality management. This situation underlines both the potential applications and the limitations of these models in varying environmental conditions.

Uncertainty of the most reliable models

Figure 6 illustrates the uncertainty of the most reliable models with key water parameters across all four seasons. The models demonstrated remarkable accuracy in predicting the maximum WQI values, with predicted maximums slightly lower than the actual observed maximums. Notably, the RF and XGB models in spring and winter exhibited the smallest error in predicting the maximum WQI, highlighting the ensemble models' ability to precisely capture scenarios of optimal water quality. The prediction of minimum WQI values showed higher accuracy in autumn and winter compared to spring and summer, likely due to the poorer water quality conditions experienced in the latter two seasons. The close alignment between the predicted and actual mean values attests to the models' effectiveness in overall

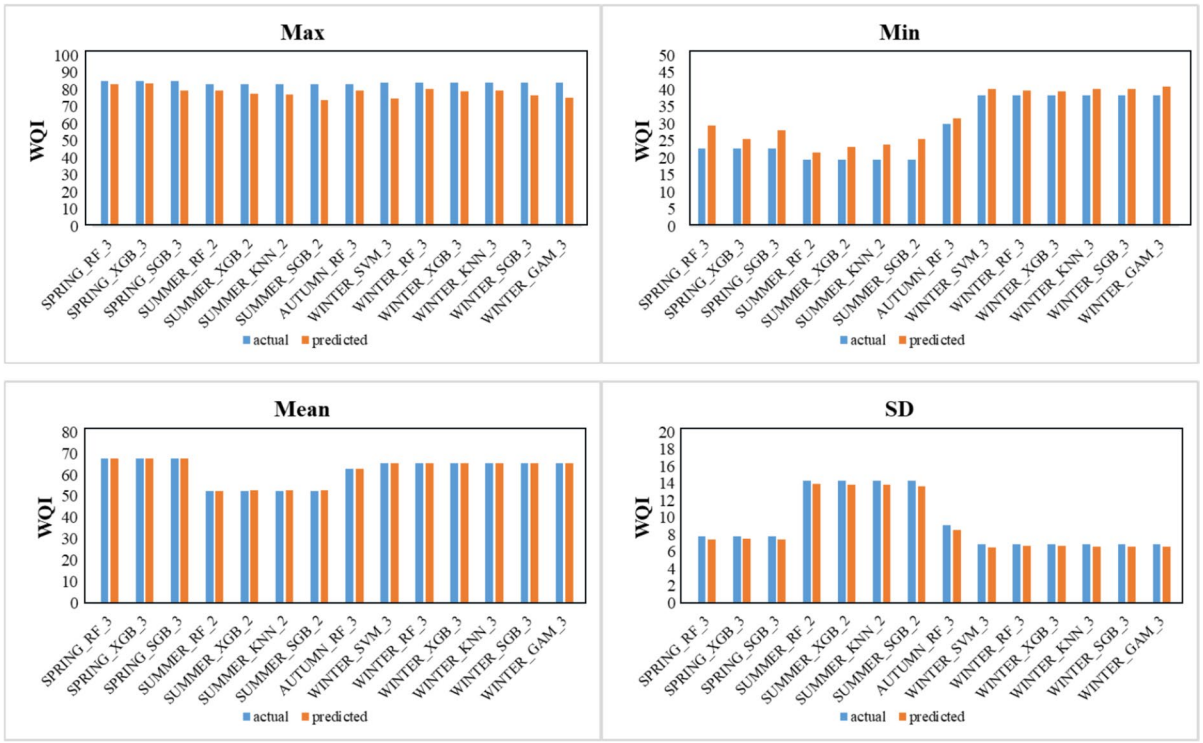


Fig. 6 The maximum (Max), minimum (Min), mean, and standard deviation (SD) of actual and predicted WQI based on key water parameters determined by the most reliable models in four seasons

water quality assessment. Throughout each season, the predicted average WQI exhibited no significant deviation from the actual average WQI, suggesting that the models can accurately reflect general water quality conditions. This makes them a powerful tool for routine water quality management and monitoring. Although the predicted standard deviations were slightly lower than the actual values, these deviations remained within an acceptable range. This indicates that while the models may not be as precise in predicting mean WQI values, their overall performance is still satisfactory.

The analysis showcases the models' exceptional ability to accurately capture various aspects of water quality, ranging from optimal conditions to general trends and adverse scenarios. This robustness across multiple metrics and seasons highlights the models' potential as powerful tools for water quality assessment and management. The comprehensive evaluation reveals remarkable stability and accuracy in the models' predictive performance across different seasons, with their exceptional precision in forecasting mean values being particularly notable. This capability is of great significance to the fields of environmental science and water management. Although the models displayed minor deviations in predicting minimum WQI values, this serves as a reminder of the need for caution and recognition of potential uncertainties when using these models to predict water quality variability. The uncertainty analysis of the most reliable models not only validates their predictive capabilities but also emphasizes their value as invaluable resources for informed decision-making and proactive measures in safeguarding water quality and promoting environmental sustainability.

Spatiotemporal variations of key water parameters

The key water parameters were identified as DO, AN, TP, and Tur based on previous results. Consequently, the spatiotemporal variations of these four indicators were evaluated. Figure 7 displays the trends and seasonal patterns in DO concentration derived using the STL method at all monitoring stations. The actual

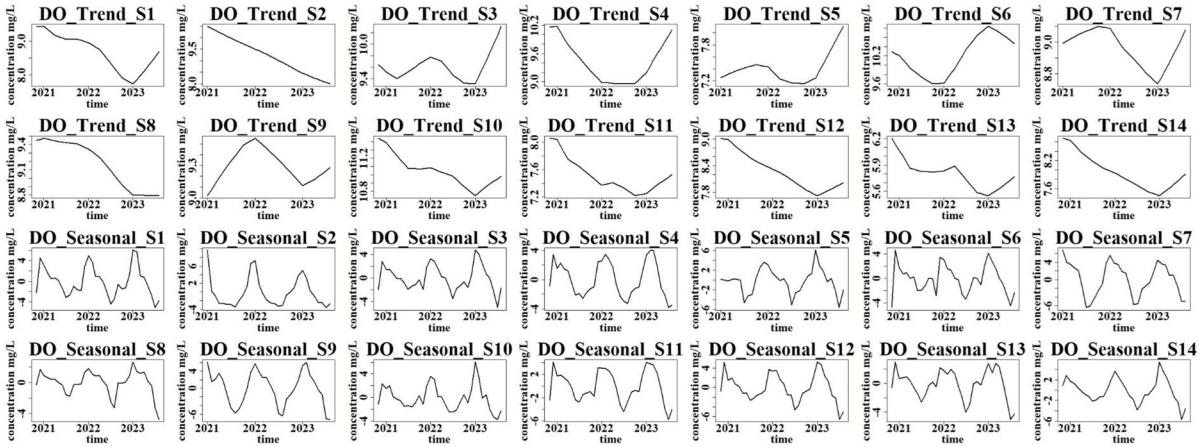


Fig. 7 Trend and seasonal components of DO concentration in all stations decomposed by the STL methods

WQI values and residual components, which reflected fluctuations unexplained by seasonal and trend factors, are illustrated in Figure S1. The trend analysis revealed noticeable increasing DO trends at stations S3 and S5, indicating improvements in water quality in those areas. However, stations S2 and S8 exhibited concerning decreasing DO trends, highlighting the need for close water quality monitoring at these sites to prevent severe oxygen depletion or anoxic conditions. At the remaining stations, the fluctuating DO trends suggested that levels were influenced by multiple factors, including seasonal temperature variations, changes in water flow dynamics, and anthropogenic activities. A common seasonal pattern emerged across nearly all stations, with peak DO concentrations occurring during winter and spring, while annual lows were observed in summer and autumn. This pattern is likely due to reduced oxygen solubility at higher summer temperatures coupled with increased biological oxygen demand. Of particular concern, stations S10, S11, S12, and S14 in southern Lianyungang City experienced abnormally low DO levels below 3 mg/L (classified as Class V water quality) from June to August 2021, which suggests the occurrence of sudden water pollution incidents.

Figure 8 illustrates the trends and seasonal variations in AN concentrations across the monitoring stations, with the raw data and residuals presented in Figure S2. An increasing trend in AN concentrations was observed at stations S1, S4, S6, and S13,

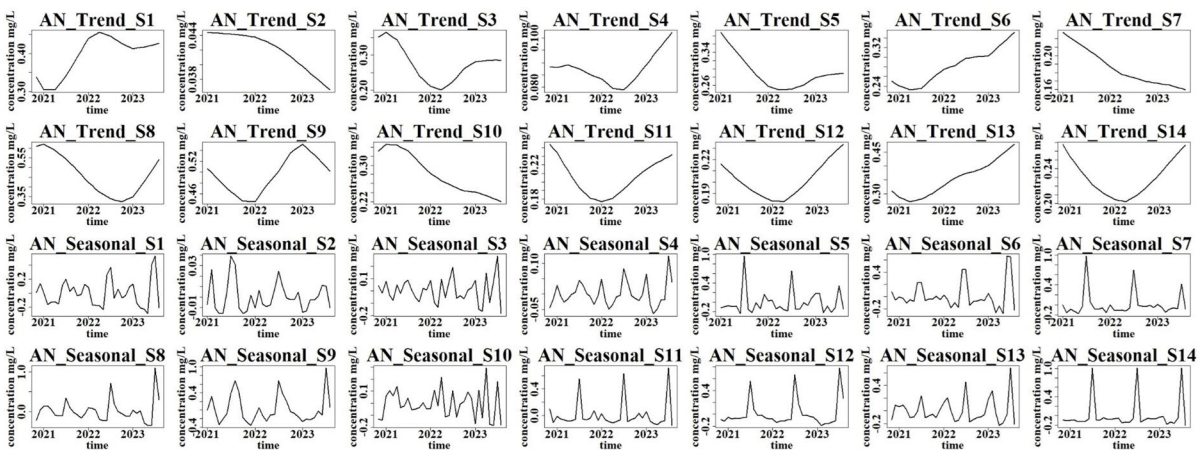


Fig. 8 Trend and seasonal components of AN concentration in all stations decomposed by the STL methods

signaling escalating AN pollution in these areas. Potential sources such as agricultural runoff, industrial effluents, or municipal sewage should be investigated in these areas, and effective emission control measures implemented to safeguard the water environment and aquatic ecosystems. In contrast, stations S2, S7, and S10 exhibited decreasing AN trends. The remaining stations displayed fluctuating AN trends, initially decreasing and then increasing, suggesting that their levels are influenced by various alternating factors. Consequently, continuous monitoring of AN concentrations and pollution sources in nearby water bodies is necessary. Lower AN levels were generally observed during winter at most stations, potentially linked to reduced agricultural activities and fertilizer application during those months. Conversely, stations like S4, S6, and S13 experienced increases in AN during spring and summer, possibly due to pollutant accumulation from intensive human activities and reduced water flows, respectively. These seasonal variations may also reflect the impact of seasonal factors such as agricultural runoff or changes in industrial activities at certain stations. For sites with significant seasonal fluctuations, flexible management strategies are recommended, including increased monitoring during peak periods and targeted water quality restoration efforts during low periods. Notably, abnormally high AN levels were recorded in July 2021 at the southern Lianyungang stations, coinciding with the timing of low dissolved oxygen events, indicating severe river pollution in that region during the summer of 2021.

The analysis in Fig. 9 explores the trends and seasonal patterns in TP concentrations, with the raw data and residuals shown in Figure S3. Stations S4 and S13 exhibited significant increasing trends in TP levels, while stations S5, S7, and S10 showed decreasing trends. The remaining sites displayed fluctuating patterns, characterized by increases followed by decreases or vice versa. Notably, station S12 underwent two distinct "U-shaped" fluctuation cycles. TP concentrations revealed pronounced seasonal variations at all stations, suggesting that TP levels were strongly influenced by factors such as precipitation patterns, watershed runoff dynamics, and agricultural activities. Station S10, located in a coastal area, experienced the most frequent and intense seasonal fluctuations in TP concentration, potentially attributed to the impacts of nearby coastal aquaculture operations (Kucuksezgin et al., 2021; Peng et al., 2022). To address these variations, enhanced monitoring efforts and tailored management strategies are recommended at this site, including the implementation of green infrastructure to mitigate phosphorus loading from stormwater runoff.

The analysis in Fig. 10 explores the trends and seasonal patterns in Tur concentrations, with the raw data and residuals displayed in Figure S4. An increasing trend in Tur levels was detected at Station S10, which may be attributed to factors such as increased river runoff, changes in precipitation patterns, and the potential impacts of riverine sediment transport and deposition processes. Stations S1, S2, and S8 exhibited fluctuating Tur patterns, while the remaining

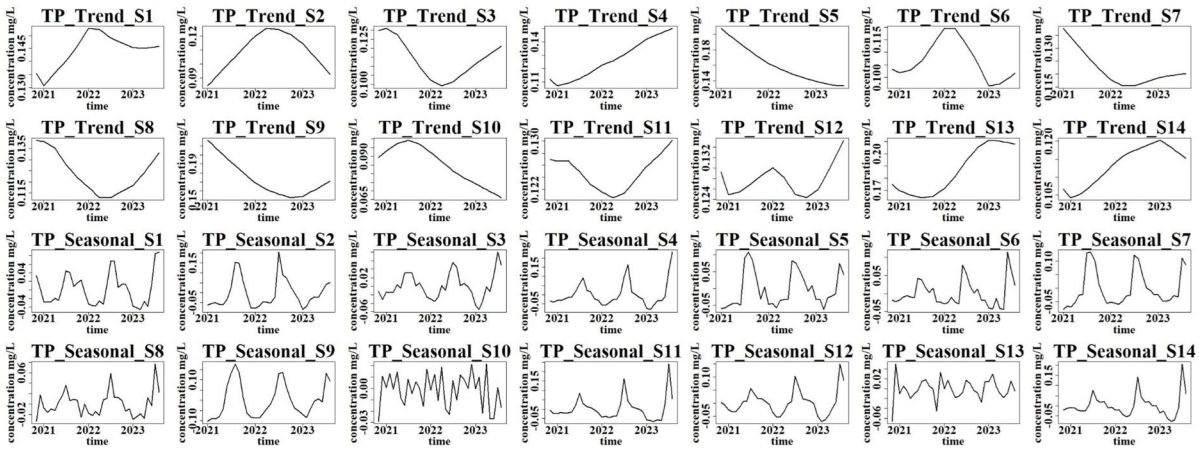


Fig. 9 Trend and seasonal components of TP concentration in all stations decomposed by the STL methods

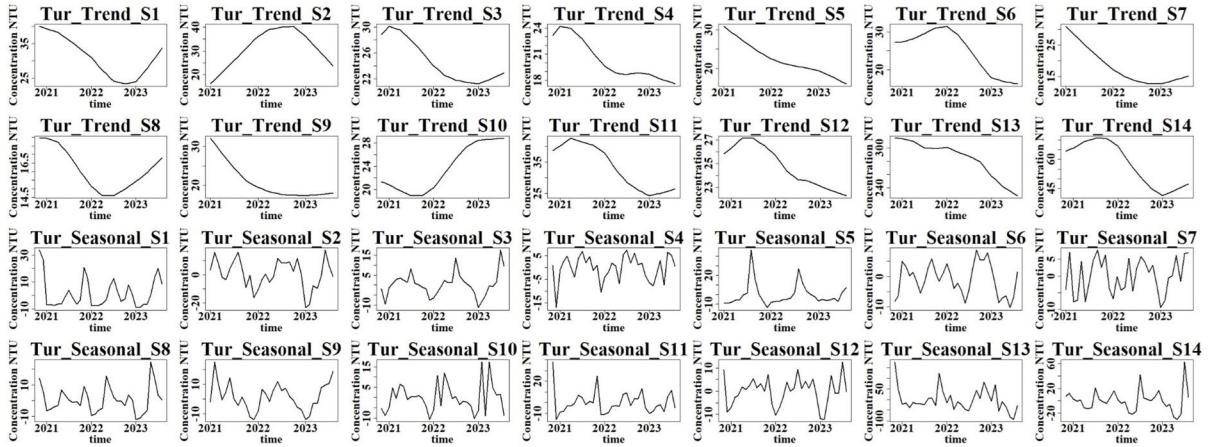


Fig. 10 Trend and seasonal components of Tur concentration in all stations decomposed by the STL methods

sites showed decreasing trends in Tur concentrations. The seasonal plots revealed distinct fluctuations in turbidity levels across all monitoring stations, which were associated with seasonal variations in precipitation and water flow dynamics. Notably, Stations S1 and S12 experienced frequent seasonal oscillations, with multiple peaks and troughs occurring throughout the year. Overall, the seasonal variations observed in turbidity appeared more pronounced compared to indicators like DO and nutrients. This finding suggests that natural hydrological conditions, including factors such as precipitation patterns and water flow regimes, exert a more substantial influence on turbidity levels than anthropogenic factors like agricultural activities and aquaculture operations.

Implications and limitations

The research and optimization efforts dedicated to these models significantly enhance the precision and efficiency of water quality monitoring systems. This advancement facilitates a more scientific and targeted approach to water quality management. Notably, accurate models are crucial in predicting trends of deteriorating water quality, providing early warnings that enable the implementation of timely remediation measures, and preventing further environmental degradation. In regions facing challenges such as scarce water resources or severe pollution, these optimized models become invaluable tools for policymaking and resource allocation. They assist decision-makers and relevant authorities in gaining a more comprehensive understanding and management of water resources, thereby supporting more informed and effective actions in environmental stewardship.

This study, despite its detailed analysis of model predictive performance for the WQI across different seasons, exhibits certain limitations. Firstly, the models demonstrate a lower degree of accuracy when predicting extreme water quality conditions, especially in cases of poor water quality. This limitation may arise from the scarcity of such conditions within the dataset or the inherent difficulty in accurately capturing the indicators associated with these states through conventional methodologies. This could potentially undermine the efficacy of the models in practical applications, such as environmental management and early warning systems. Although the study considers the influence of seasonality on water quality predictions, it does not fully capture the complex dynamic changes impacting the biological, chemical, and physical characteristics of water bodies within the confines of the models.

Conclusions

In the research, we aimed to determine seasonal key water parameters and the most reliable models to predict WQI based on water quality dataset from 2020 to 2023 in a coastal city in China, by which maximum precision could be gained by minimum input variables. Key findings are summarized as follows:

(1) Nutrient concentrations were higher at inland stations compared to coastal stations, whereas EC and Tur displayed the opposite patterns. Water quality conditions were better in spring and winter, with the majority of records classified as Good or Medium grades. However, conditions deteriorated in summer, marked by an increase in the proportion of Low and Bad grades.
(2) The four key water parameters identified were nutrients (AN and TP), DO, and Tur. All of these parameters exhibited seasonal variations, influenced by dynamic changes in natural conditions and human activities. Trends in these parameters varied across monitoring stations, displaying increasing, decreasing, or exhibiting "U-shaped" fluctuations. Generally, turbidity concentrations decreased, but nutrient pollution worsened at coastal stations.
(3) The performance of most models significantly improved with an increase in the number of input variables, except for certain models such as SOM and KNN. Tree-based boosting methods, including XGB and SGB, along with the RF model, demonstrated high predictive performance. Other models showed varying performance across different seasons, achieving the highest accuracy in winter.
(4) The RF and XGB models showed relatively consistent and superior performance with key water parameters in spring and winter, whereas the KNN and RF models performed better in summer. The predictive accuracy for Medium and Low water quality grades with key water parameters was perfect across all four seasons. For Good water quality grades, the models exhibited higher predictive accuracy in spring and winter (over 80%) compared to summer and autumn (70%). However, the predictive accuracy for Bad grades generally remained below 50%.

Reliable WQI prediction models are crucial for assessing water quality, particularly when data availability is restricted. These models can aid in the efficient and effective management of water resources, for example, by offering early warnings that facilitate the implementation of timely remediation measures and help prevent further environmental degradation. However, this study also reveals certain limitations. For instance, the models show reduced accuracy when predicting poor water quality. This limitation may arise from the infrequency of such conditions within the dataset or the inherent challenges in accurately capturing the indicators associated with these states using conventional methodologies.

Author contribution Yuming Mo: Data curation, Funding acquisition, Methodology, Software, Validation, Visualization, Writing—original draft; Jing Xu: Funding acquisition, Methodology, Supervision, Writing—review &editing; Chanjuan Liu: Methodology, Writing—review &editing; Jinran Wu: Methodology, Validation, Writing—review &editing; Dong Chen: Investigation, Writing—review & editing.

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Data availability Data will be available from the corresponding author on reasonable request (xujing7503@yzu.edu.cn).

Declarations

Competing interests The authors declare no competing interests.

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